

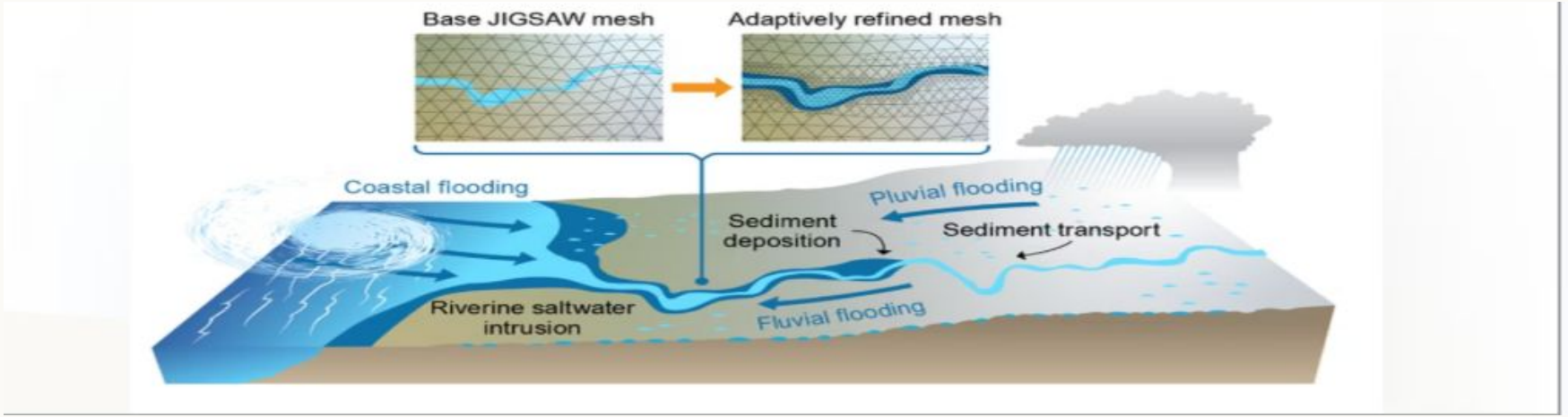
Comparison of Roe, HLL, and HLLC Riemann Solvers in RDycore

Accuracy, diffusion, and GPU considerations for shallow-water simulations

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41st Colorado Conference on Iterative and Multigrid Methods
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Motivation: Compound Flooding Simulations

Real flood events combine coastal, riverine, and rainfall-driven processes.



Why this is computationally hard

Large domains and long simulation times
Complex topography and frictional effects
Shocks, wetting/drying, and moving fronts
Operational use requires efficient execution

Riemann solvers influence both accuracy and cost.

Why performance matters

- Adaptive meshes refine narrow stream channels
- Small cells impose restrictive CFL limits
- Long-duration simulations (100+ years) require billions of updates
- Strong scaling and solver efficiency become critical

RDycore Framework

A finite-volume compound flooding code with PETSc/libCEED infrastructure.

RDycore connects a shallow-water flood model to modern HPC infrastructure.

Numerical framework

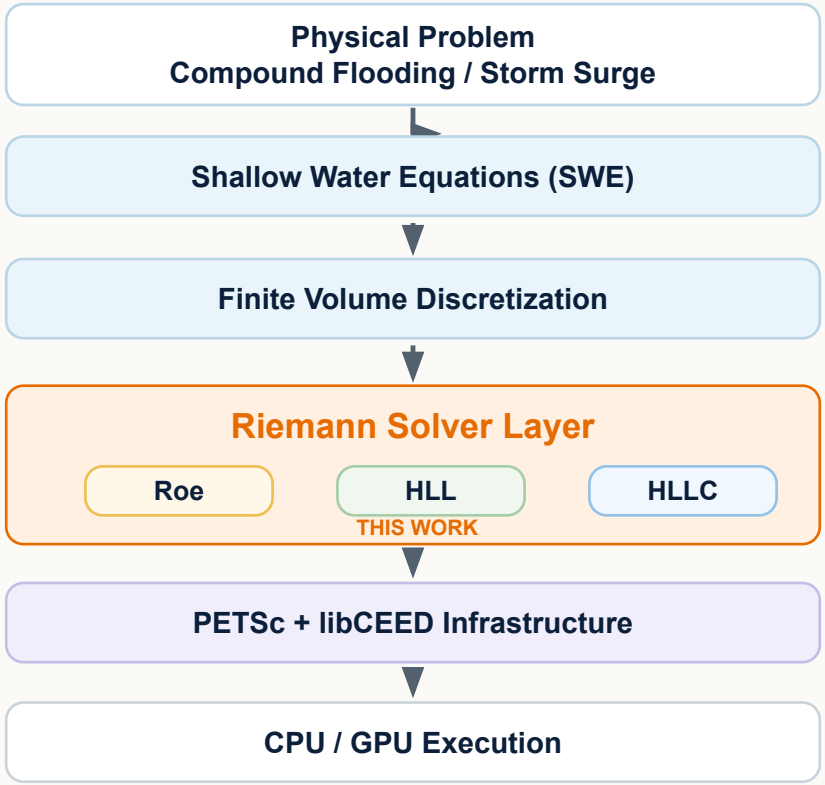
Finite-volume discretization
SWE over complex topography
Modular flux/source kernels

HPC infrastructure

PETSc parallelism
libCEED operator abstraction
CPU and GPU backends

This work

Implement HLL solver
Implement HLLC solver
Compare against Roe



RDycore docs: Theory Guide overview and SWE spatial discretization.

Governing Model: 2D Shallow-Water Equations

RDycore solves the shallow-water equations in conservative balance-law form.

$$\frac{\partial U}{\partial t} + \frac{\partial F(U)}{\partial x} + \frac{\partial G(U)}{\partial y} = S(U, z_b)$$

Conserved Variables

$$U = \begin{bmatrix} h \\ hu \\ hv \end{bmatrix}$$

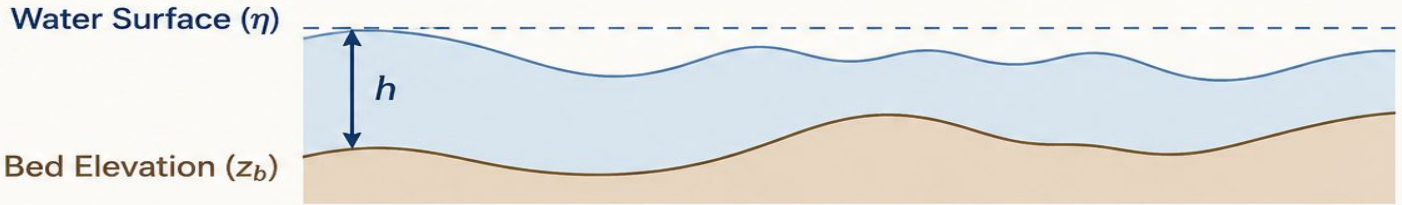
h : water depth

hu : x-momentum

hv : y-momentum

Source Terms

- Bed slope
- Manning friction
- External forcing
- Wetting / drying effects



$$\eta = h + z_b$$

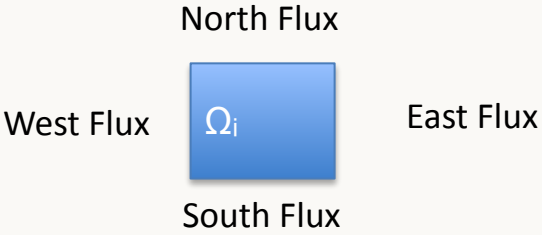
Key Idea: Flood simulations are governed by balance laws that combine conservation with topographic and frictional source terms.

Finite-Volume Discretization

The update is driven by fluxes crossing the faces of each cell.

Cell-average update

$$Q_i^{n+1} = Q_i^n - \frac{\Delta t}{|\Omega_i|} \sum_j F_{ij} A_{ij}$$



Conservation is enforced through fluxes crossing cell faces.

Why finite volume?

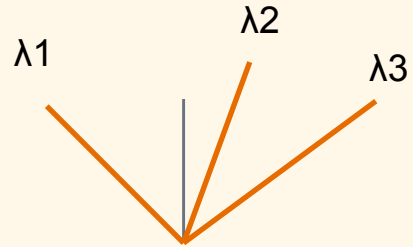
- Local conservation
- Works naturally on unstructured meshes
- Robust for discontinuous solutions

Key Idea: Cell States → Numerical Flux → Finite-Volume Update

Approximate Riemann Solvers Compared

The solvers differ in wave structure, numerical diffusion, and implementation cost.

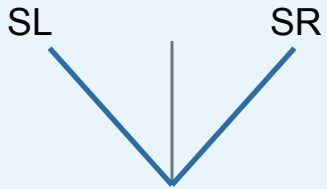
Roe



Linearized characteristic-wave model

Linearized Jacobian
Sharp resolution
Requires entropy fix
More algebra / eigenstructure

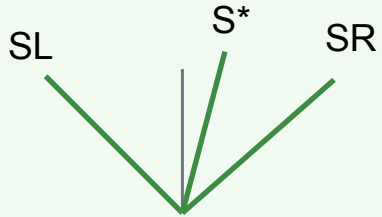
HLL



2-shock approximation

Two-shock approximation
Robust and simple
No contact wave
More numerical diffusion

HLLC



adds contact wave

HLL + contact wave
Reduced diffusion vs HLL
Moderate complexity
Sharper intermediate states

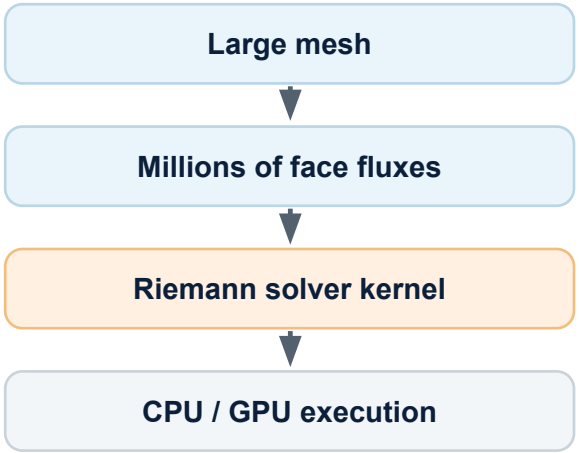
Question: how do these differences show up in accuracy, diffusion, stability, and runtime?

HLL uses a two-shock approximation; HLLC restores the contact wave. Roe requires an entropy fix.

GPU Considerations

Solver structure can affect performance beyond raw floating-point count.

Why solver cost matters



Implementation-sensitive features

- **Branching** — different wave-speed cases
- **Arithmetic intensity** — more operations per face
- **Memory access** — states, geometry, sources
- **Kernel efficiency** — backend and data layout

Accuracy and GPU behavior must be assessed together.

GPU emphasis follows the project abstract: execution behavior depends on solver formulation and branching.

Objectives and Evaluation Plan

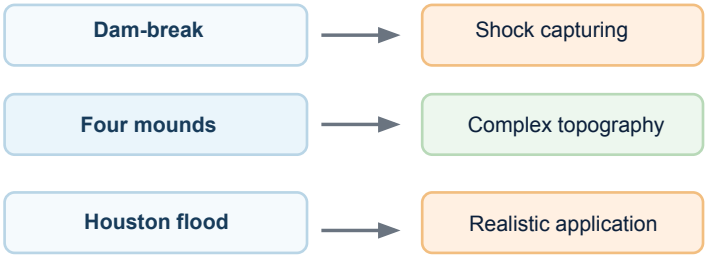
The introduction leads directly into the numerical results.

Research questions

How do Roe, HLL, and HLLC compare in accuracy?
How much numerical diffusion is introduced?
How do solvers behave over variable topography?
What are the implications for GPU execution?

Implement HLL/HLLC → compare with Roe

Intro-to-results roadmap



Next: numerical results across idealized and realistic flooding benchmarks.

Square Dam-Break Setup

Flat-bed benchmark for shock-capturing and solver comparison.

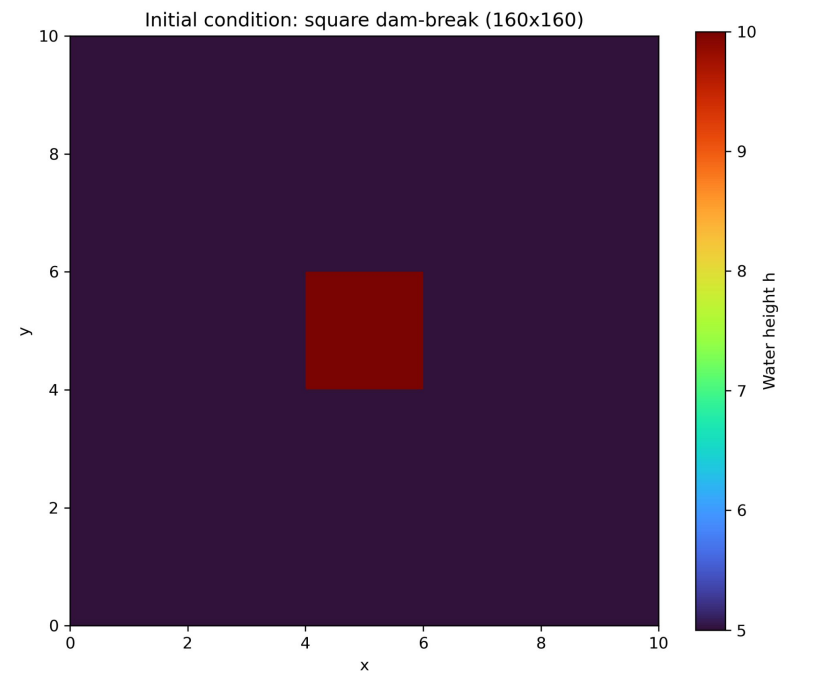
Benchmark Configuration

- Domain: 1 km × 1 km
- Flat bed bathymetry (zb = 0)
- Initial velocity: u = v = 0
- Reflective wall boundaries
- Mesh resolutions:
  40×40, 80×80, 160×160, 320×320

Initial Water Depth

h = 10 m inside the central square

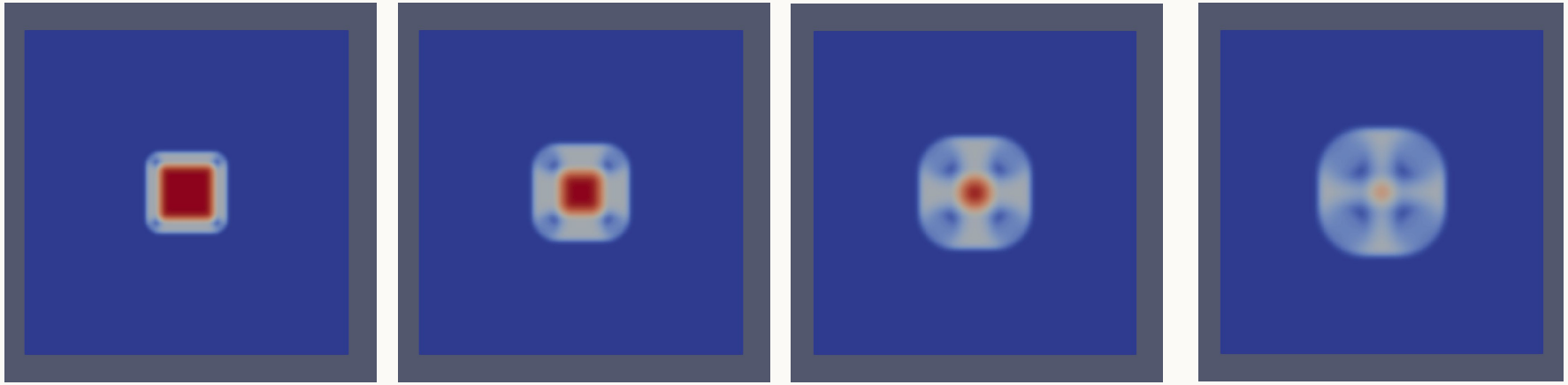
h = 5 m elsewhere



KEY INSIGHT

A simple 2D benchmark that generates shocks,rarefactions, and multidimensional wave interactions without source-term effects.

Temporal Evolution of the Square Dam-Break Solution



Time steps:

t = 300 s

t = 700 s

t = 1100 s

t = 1500 s

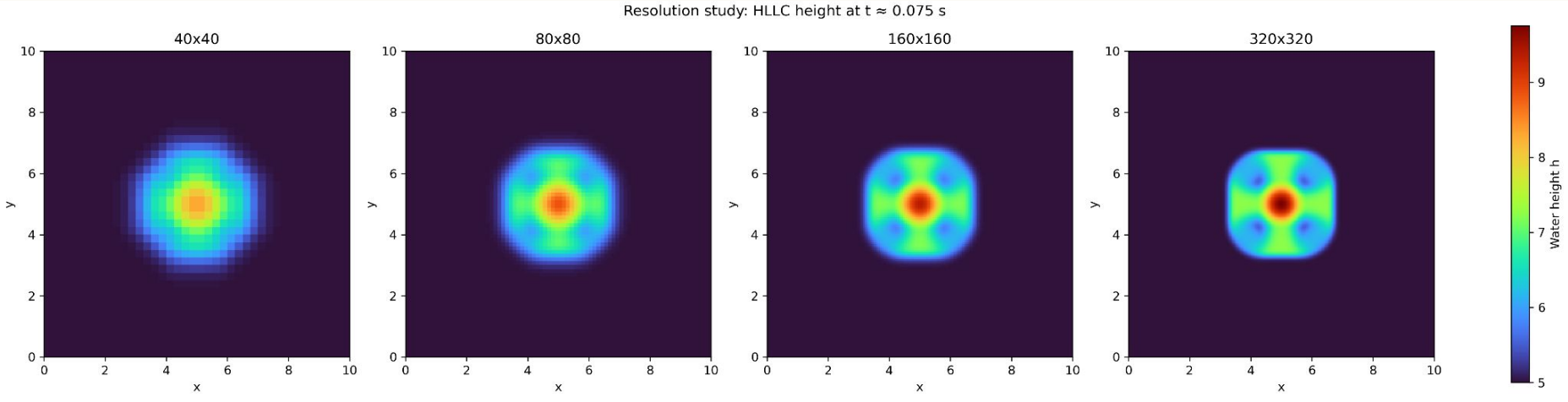
Observations

- The initial discontinuity generates outward-propagating waves.
- Rarefaction and shock structures emerge rapidly.
- Multidimensional interactions develop near the corners.
- The benchmark produces genuinely two-dimensional flow features.

Key Insight

The square dam-break generates rich two-dimensional wave structures suitable for comparing numerical diffusion and shock-capturing behavior.

Resolution Study



Increasing Resolution

40×40

80×80

160×160

320×320

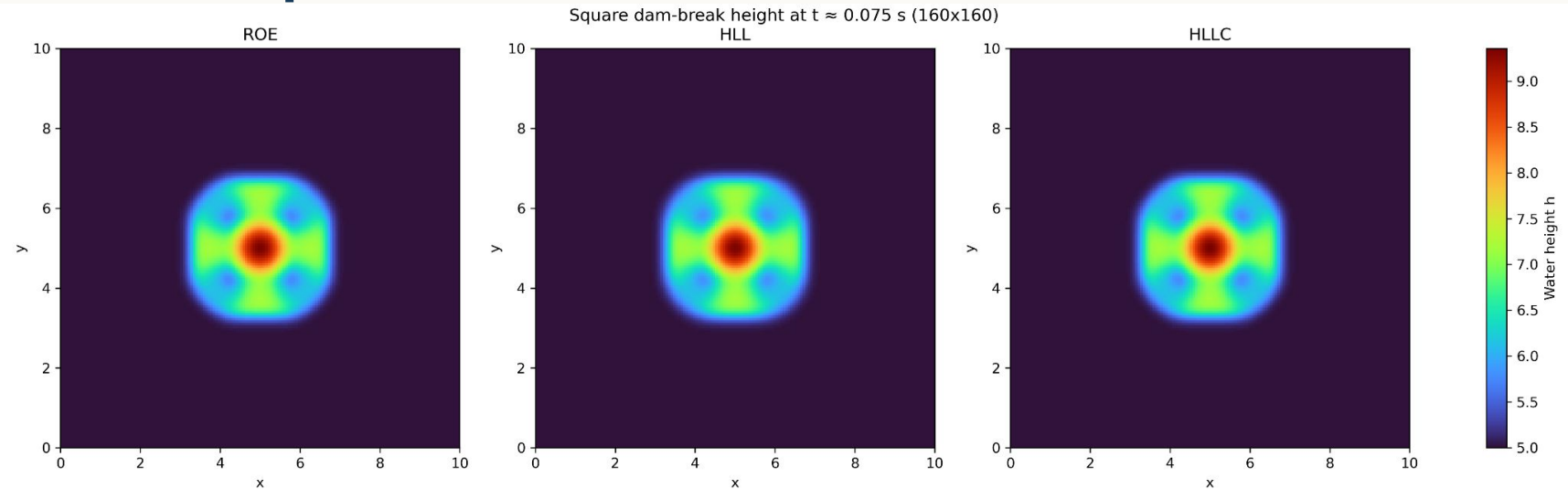
Observations

- Wave fronts become sharper
- Numerical diffusion decreases
- Fine meshes resolve additional flow structure
- Physical domain remains unchanged

Key Insight

Spatial discretization errors remain significant even before considering solver differences.

Solver Comparison at Fixed Resolution



Roe

- Sharp wave resolution
- Lowest diffusion

HLL

- Robust and simple
- Most diffusive solution

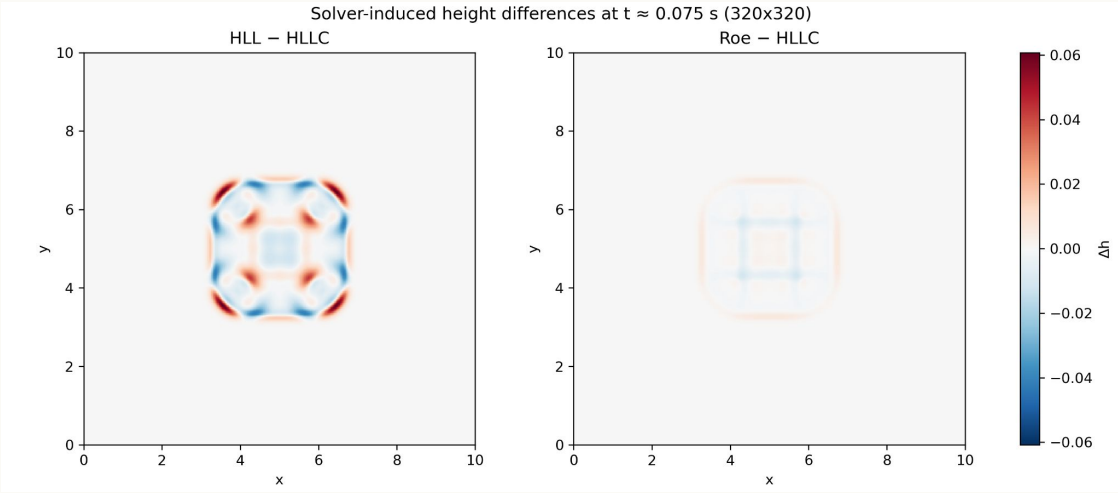
HLLC

- Restores the middle wave
- Closely matches Roe

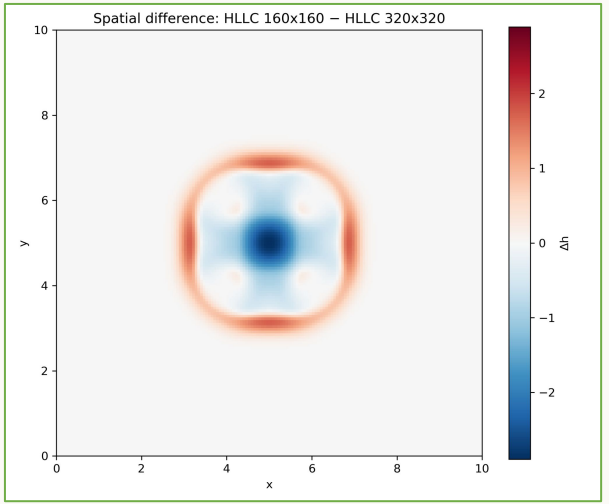
Key Observation: Roe and HLLC produce nearly identical solutions, while HLL introduces additional diffusion.

Solver Error vs Spatial Error

Max |Δh| ≈ 0.06 m



Max |Δh| ≈ 3 m



Solver Difference

HLL − HLLC

Localized near shock fronts

Spatial Difference

HLLC 160² − HLLC 320²

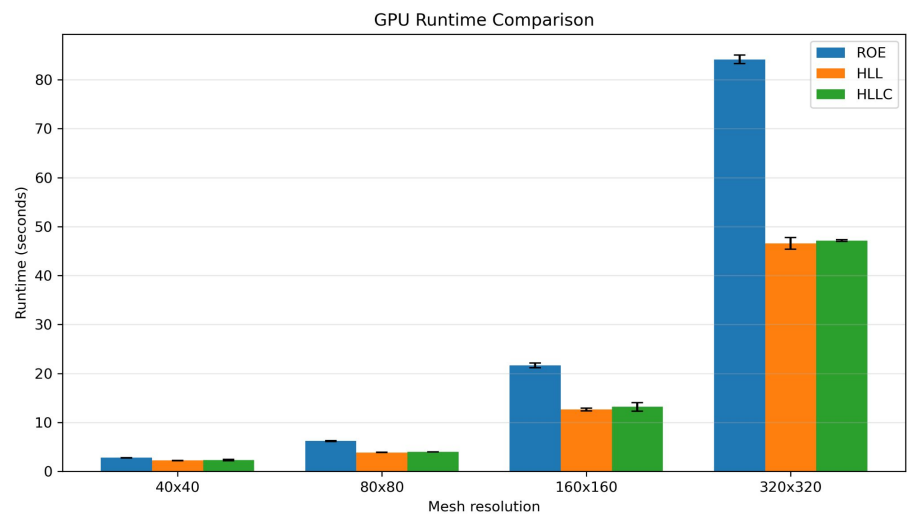
Effect of mesh refinement

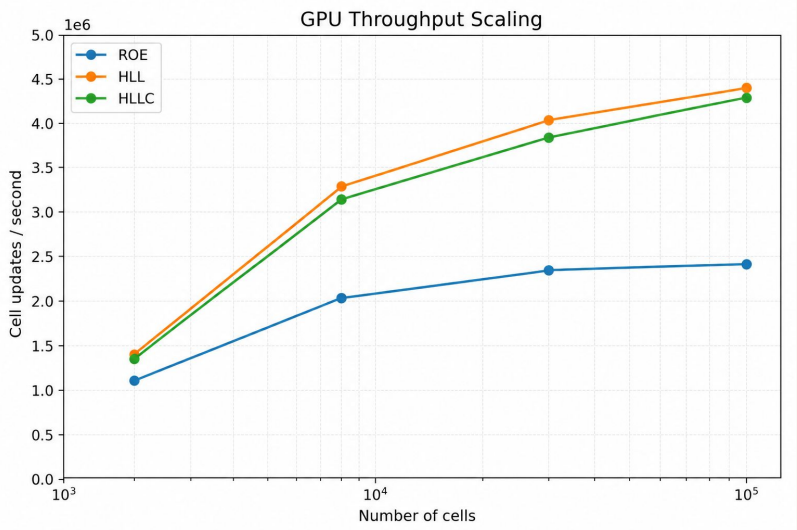
Observation

- Differences are concentrated near shock fronts
- Mesh refinement produces substantially larger changes
- Maximum spatial differences are roughly 50× larger than solver differences

Key Observation: Spatial discretization errors are substantially larger than solver-induced differences. Mesh refinement has a stronger impact on the solution than the choice of Riemann solver.

Accuracy–Cost Tradeoff





Hardware / Backend:
NVIDIA GTX 1050, 4 GB VRAM; CUDA backend via PETSc/libCEED; WSL Ubuntu

Speedup Relative to Roe

HLL : 1.81×

HLLC : 1.78×

Observation

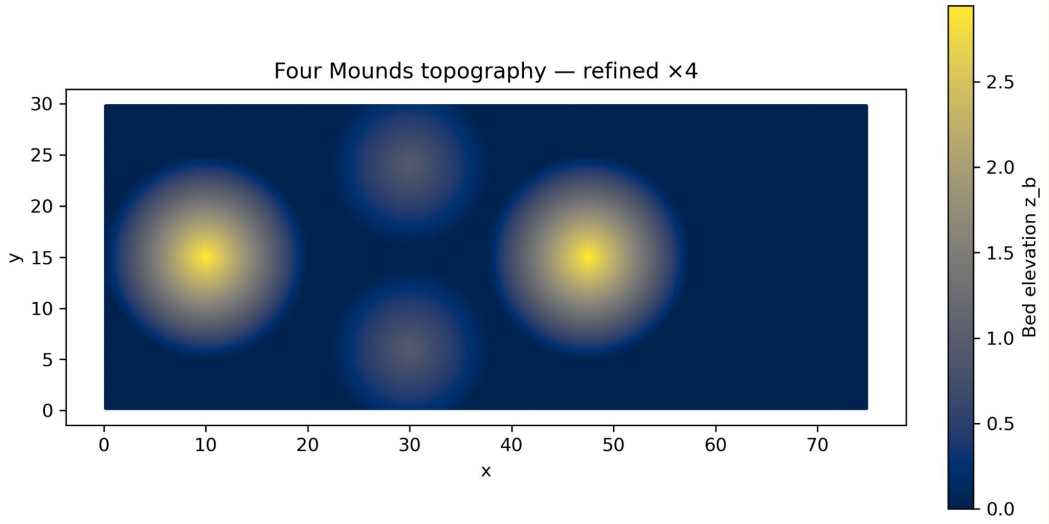
- HLL is the fastest solver
- HLLC is only marginally slower
- Both significantly outperform Roe

Key Observation: HLLC retains Roe-like accuracy while achieving nearly the same cost as HLL. HLL remains slightly cheaper, but the cost difference between HLL and HLLC is negligible.

Four Mounds Benchmark Setup

Four Mounds Benchmark: Complex Topography

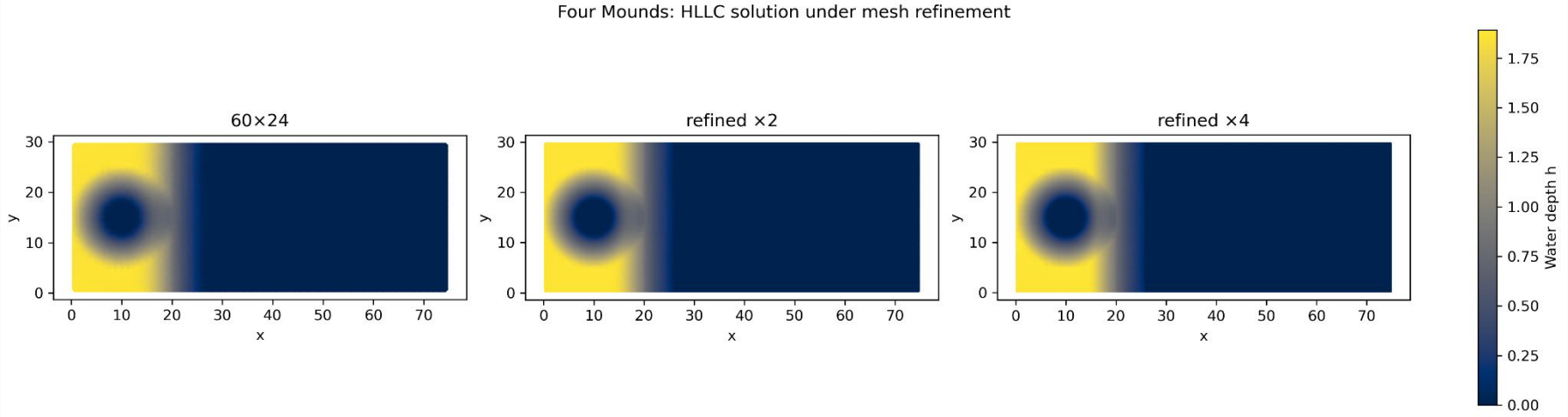
- 2D shallow-water test over non-flat bed elevation
- Four mound obstacles generate transverse flow structure
- Compare Roe, HLL, and HLLC under mesh refinement
- Resolutions: 60×24, refined ×2, refined ×4



Mesh Refinement / Solution Structure

Solution Structure Under Mesh Refinement

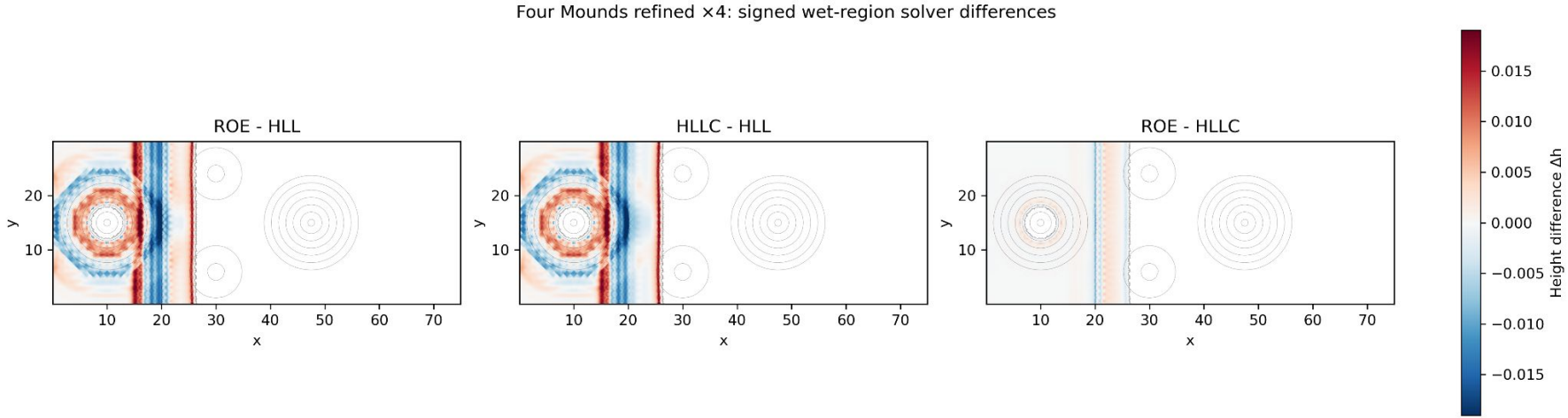
Water depth at t = 1.6 s



- HLLC solution shown at three spatial resolutions
- Refinement sharpens the wet/dry front and mound interaction
- Overall flow structure remains consistent across meshes
- Consistent flow structure under refinement suggests regular convergence.

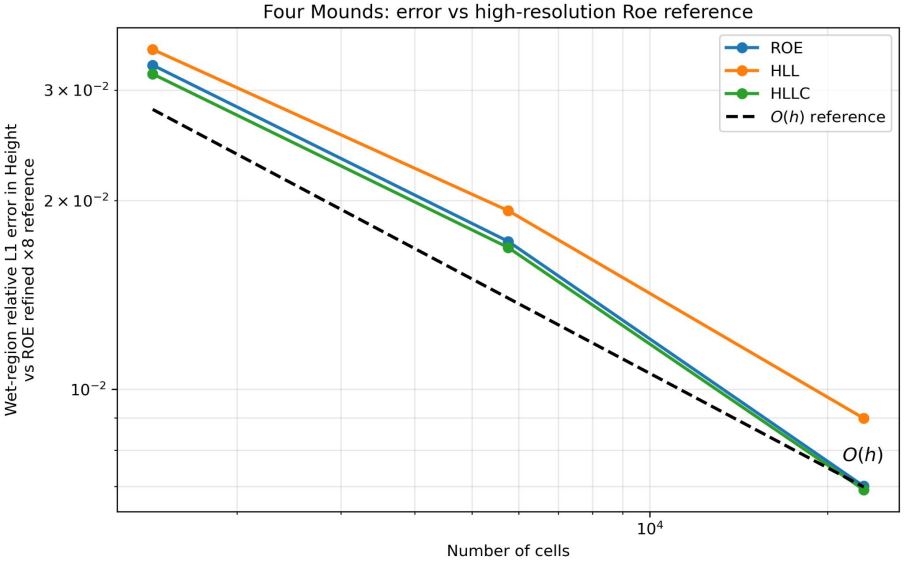
Solver Comparison at Finest Resolution

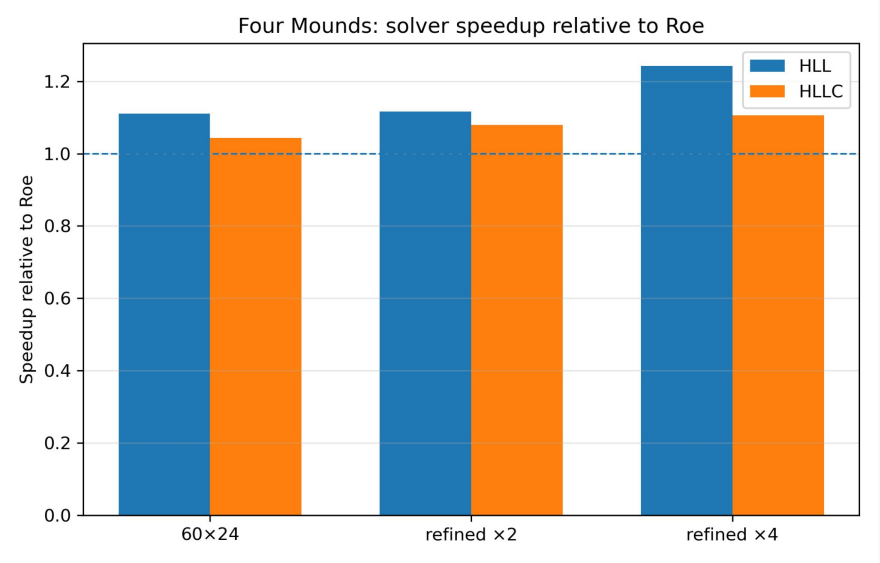
Solver Differences at Refined ×4 Resolution



- Roe and HLLC are much closer to each other than either is to HLL
- Differences are localized near the wet/dry front
- HLLC preserves Roe-like solution quality on complex topography

Accuracy–Cost Tradeoff: Error vs Reference



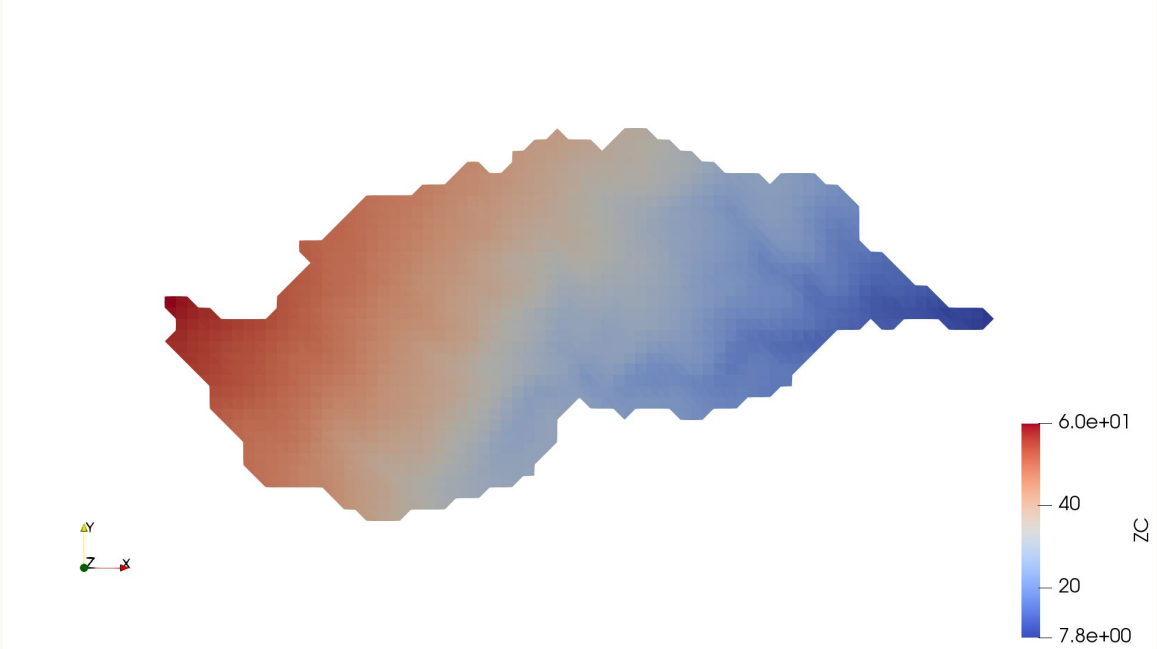


- Errors are measured against a high-resolution Roe reference solution.
- All solvers show similar convergence trends under mesh refinement.
- HLL has a larger error constant, consistent with higher numerical diffusion.
- HLLC remains close to Roe while preserving a lower implementation cost.

Houston 1 km Setup

Houston 1 km realistic topography test

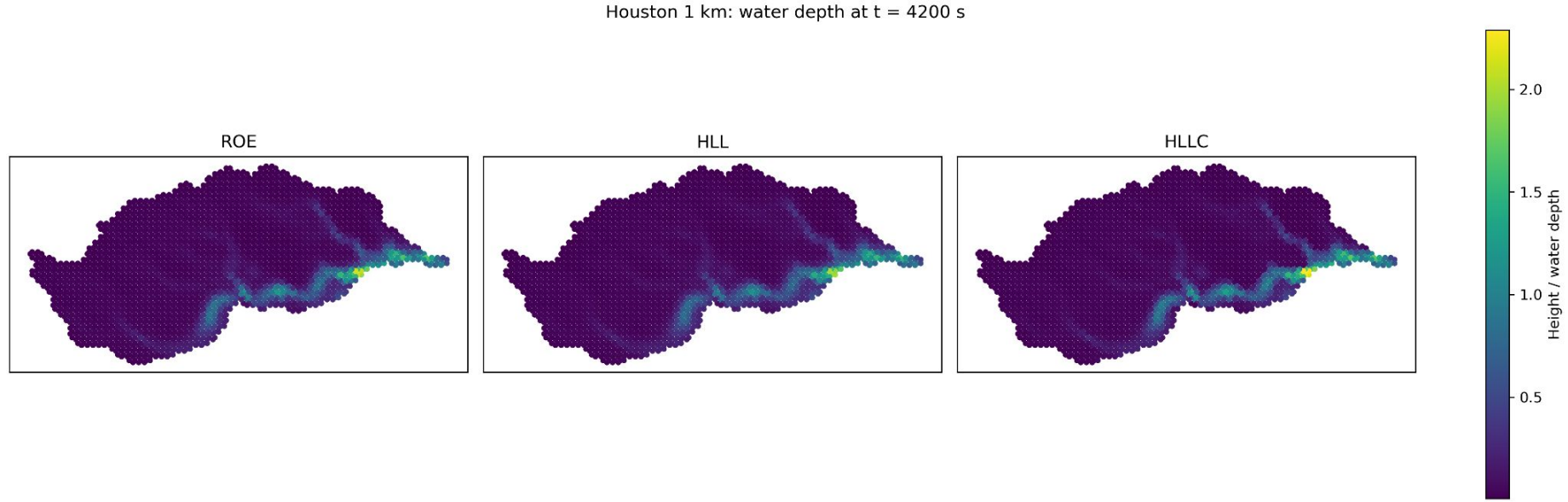
- 2746 finite-volume cells
- Real topographic elevation
- Manning coefficient n = 0.015
- Simulation time: 4200 s
- Comparison of Roe, HLL, and HLLC solvers



KEY INSIGHT

This test evaluates solver behavior on a realistic large-scale flooding domain.

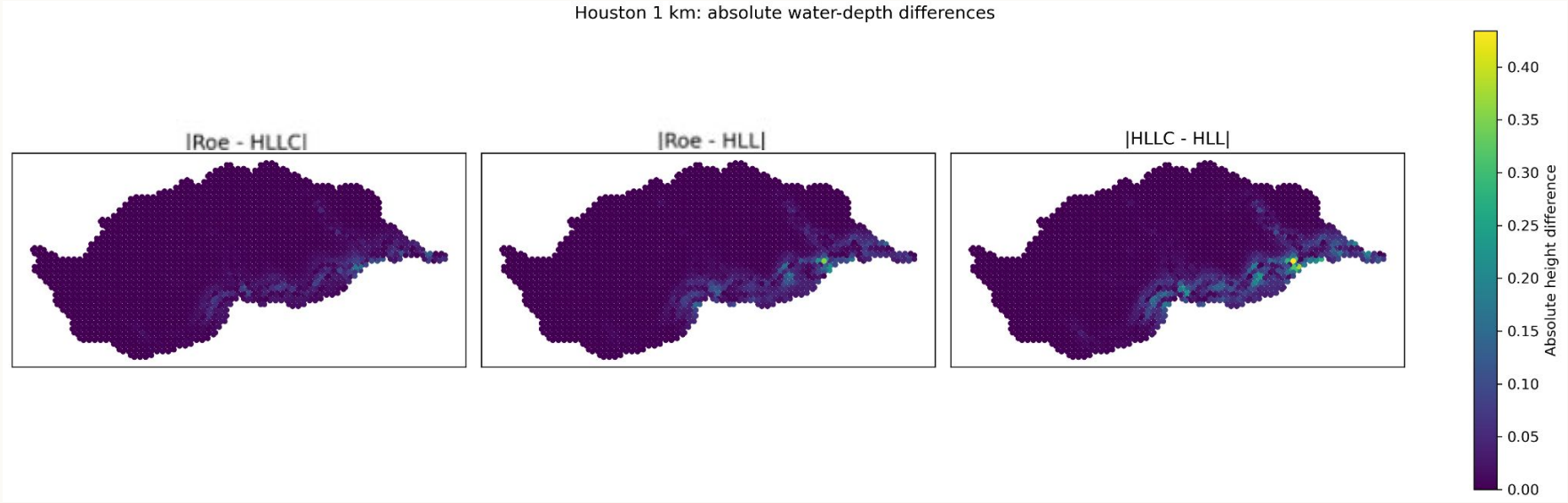
Solution Comparison



Key Observation:

- All three solvers predict nearly identical flood patterns.
- Major flow pathways are preserved.
- Differences are visually small at this resolution.

Localized Solver Differences



Key Observations

- Differences are localized near active flow channels.
- Maximum discrepancies remain small relative to local water depth.
- Roe and HLLC remain substantially closer to each other than either is to HLL.
- Major flood pathways are preserved by all three solvers.

Key Findings

Accuracy

- Roe and HLLC produce nearly identical solutions.
- HLL introduces additional diffusion.
- Solver differences decrease under mesh refinement.

Realistic Applications

- Houston benchmark shows nearly identical flood predictions.
- Major flow pathways are preserved.

Performance

- HLL and HLLC are ~1.8× faster than Roe.
- Runtime difference between HLL and HLLC is negligible (~1%).

Conclusions and Future Work

Conclusions

Roe

- Sharpest wave resolution
- Lowest numerical diffusion
- Most expensive solver

HLL

- Simplest formulation
- Fastest solver
- Highest diffusion

HLLC

- Roe-like accuracy
- HLL-like cost
- Best accuracy/cost tradeoff

Maintainability

HLL/HLLC Advantages

- Simpler wave structure
- No full eigenstructure decomposition
- Easier to maintain
- Easier to extend

Applications:

- Salinity transport
- Sediment transport
- Density-dependent flows

Future Work

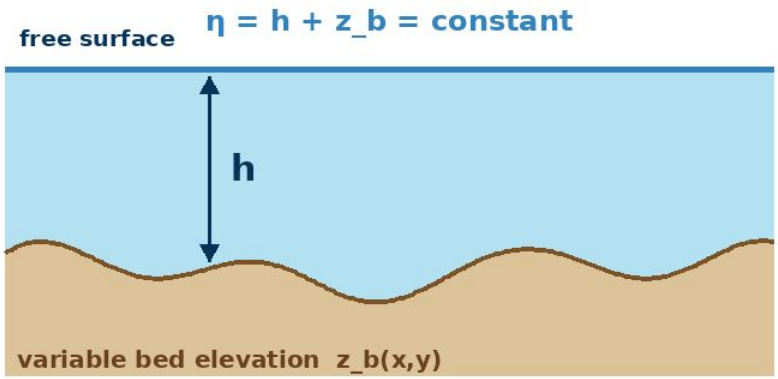
- Full GPU benchmarking
- Additional active scalar fields

Thank you

Topography and Well-Balancing

A well-balanced discretization preserves physically steady states over variable topography.

Lake-at-rest steady state



$u = v = 0$

Why it matters

- Preserves lake-at-rest equilibrium
- Prevents spurious numerical waves
- Balances flux gradients with bed-slope source terms
- Critical for simulations over variable topography

well-balanced cancellation

$$\nabla(\tfrac{1}{2} g h^2) + g h \nabla z_b \approx 0$$

flux gradient — bed-slope source

Key Idea: A well-balanced scheme maintains $u = v = 0$ and $\eta = h + z_b = \text{constant}$ without artificial waves.

Lake-at-rest condition: flux/source cancellation over variable topography.